



Modify curve end

Move one end of a curve, chain or wire to a new point and blend the change back -- hold part of the curve exactly, keep G1/G2 there, and match the end to a reference.

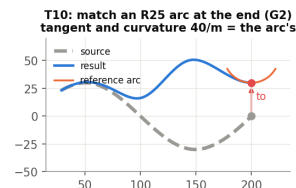
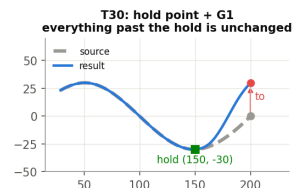
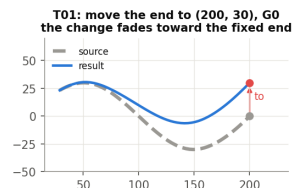


What it does

Moving an end of a network curve usually means redrawing it. Modify curve end edits the curve's own control points: the end goes where you ask, the change fades back, and whatever you hold stays exactly as it was.

Use it when

- Stretch a curve to a new corner of the network
- Keep most of a curve untouched
- Land an end tangent (or curvature-continuous) to an edge, face or mate connector

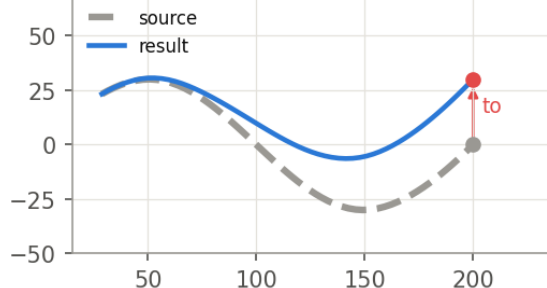


T01 move, T30 hold, T10 match an arc (G2).



How it works

T01: move the end to (200, 30), G0
the change fades toward the fixed end



Control points, not a refit

Free control points move by a share of the offset that fades by arc length; the last moves fully.

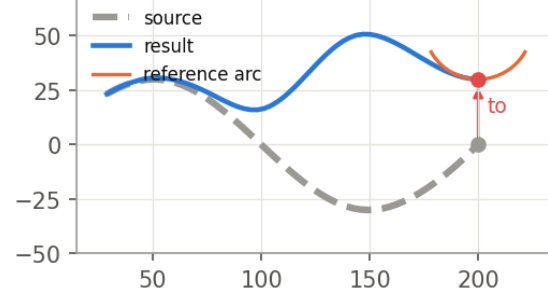
T30: hold point + G1
everything past the hold is unchanged



Exact hold

Knots at the hold make the held part depend only on locked control points; G1 / G2 lock one / two more.

T10: match an R25 arc at the end (G2)
tangent and curvature $40/m = \text{the arc's}$



Closed-form end

Tangent (and curvature) of an edge, face or mate connector set directly on the last control points.



The dialog



#	Parameter	What it means
1	Edges or wire to modify	An edge, a chain in any order, or a wire; open, unbranched.
2	Modified end	The end to move (empty = the chain's end).
3	To point	Where it goes (empty = stays / snaps onto the reference).
4	Hold part of the curve	Hold by Point or Distance (arc length from the modified end).
5	Parameters	Transition type; continuity at the fixed end / hold (G0 / G1 / G2); end reference.
6	Project onto surface?	+ face: only the part after the hold is projected (G1 kept at the hold).
7	Debug, Details	Degree, tolerance, max control points of a merged chain, sampling multiple, print.



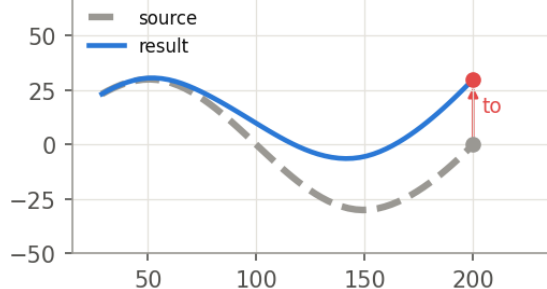
End reference (in Parameters)

Parameter	What it means
Endpoint continuity ref.	An edge, a face or a mate connector.
Opposite direction	Reverse the end tangent.
Match	G0 / G1 / G2 at the moved end.
End curvature	(G2) Match the reference / Zero / Radius.
End radius	100 mm; bends toward the connector's X, else the edge's normal, else the curve's own.
Offset frame	No effect any more (kept for saved features).

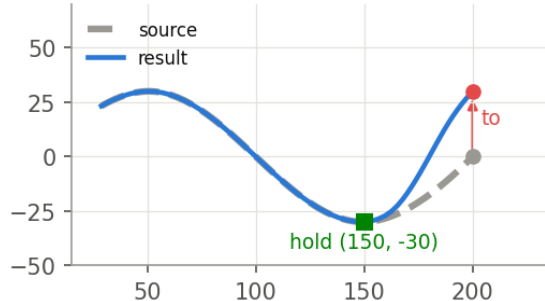


Example: hold point (T30)

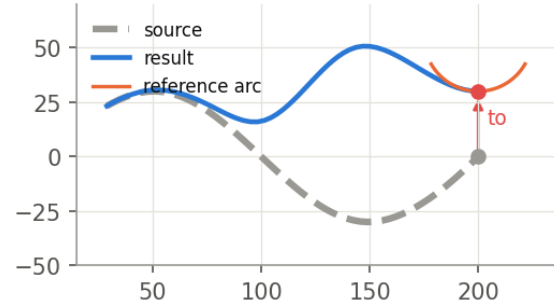
T01: move the end to (200, 30), G0
the change fades toward the fixed end



T30: hold point + G1
everything past the hold is unchanged



T10: match an R25 arc at the end (G2)
tangent and curvature 40/m = the arc's



End (200, 0) -> (200, 30); hold at (150, -30), G1

Everything from the fixed end to the hold is unchanged (under 1 micron); the tangent at the hold is kept. T10: matching an R25 arc gives end curvature 40/m, as the arc.



Outputs and messages

Published for Extract variables

Key	What it holds
output	The modified curve (one wire).
movedVertex	The end at the To point.
holdVertex	The hold point, or the fixed end when there is no hold.

Messages

	Message / what to do
Info	Notes: no Modified end picked, hold has no effect, G0 hold leaves a corner, reference far away, handle shortened, projection keeps G1 only One line.
Error	branched / several chains / closed chain Select one open chain.
Error	hold point missing or at the modified end; hold distance longer than the curve



Tips and limits



Hold instead of G2 everywhere

A hold keeps the part you care about exactly; blend only the rest.



Mate connector reference

Z = the end tangent, X = the bend side; set a radius for exact end curvature (T13).



Projection

Keeps the end tangent only if it lies in the face; end curvature cannot be kept.



Where it lives

Document	<code>gordonSurface (d0950ed72a894fbe35c27d43)</code>
Tab	<code>modifyCurveEnd</code>
Tests	Gordon tools tests: T01-T40 (all pass)
Examples	Test Part Studio 1: Modify curve end 1
Explainer	<code>docs/explainers/gordon_surface/gordon_surface_explained.md</code> , section 2.4
Related	Gordon Surface, Interior curves, Scaled Curve, Modify curve end, Pull surface, Simplify surface